

Charles Chow

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SUMMARY

Mechatronics engineer focused on robotic hardware, electromechanical motion systems, embedded sensor firmware, PCB-based controls, CAD, fixtures, BOMs, and hardware/software debugging. Mechanical foundation with CS/ECE graduate training across controls, embedded systems, scripting, and prototype validation.

SKILLS

Mechanical / CAD	SolidWorks, NX, Fusion 360, Ansys, FEA, GD&T, BOMs, engineering drawings, 3D prototyping
Electromechanical	Motor selection, PID control, sensor integration, PCB design, wiring harnesses, electromechanical assembly
Embedded / Firmware	C/C++, Zephyr RTOS, nRF52, BLE/GATT, UART, I2C, SPI, Nordic SDK, SEGGER RTT, Linux
Software	Python, ROS2, Nav2, Gazebo, WebSockets, PostgreSQL, Git, Bash scripting
Electrical / Test	Schematics, wiring diagrams, multimeter, oscilloscope debugging, data logging, failure analysis

RELEVANT EXPERIENCE

Embedded Software Intern, NeuroLeap May 2025 – Dec 2025

- Designed embedded C firmware on Nordic nRF52840 for 20 Hz multi-channel BLE sensor streaming, supporting validation, algorithm development, and QA workflows across 5+ engineers.
- Tuned BLE link parameters to reduce packet loss from 9.2% to 1.8%, cutting recalibration time by 60% and enabling uninterrupted 8-hour test sessions.

Mechanical Engineer, Keystone Tower Systems Jun 2022 – Mar 2023

- Designed electromechanical motion subsystems for a \$2.3M automated welding system incorporating stepper, servo, and DC motors; collaborated with controls engineers on actuator specs and motion sequencing, achieving 15% faster cycle times.
- Built kinematic layouts enforcing +/-0.5 mm positioning tolerances across 6 configurations; resolved 3 collision scenarios through simulation-driven design review.
- Owned mechanical, controls, and field integration end-to-end; produced 2 technical documentation packages.
- Ran Ansys structural, thermal, and motion FEA across multiple subsystem designs; used simulation results to drive design decisions on material selection, geometry changes, and load path optimization.

Mechanical Engineer, Engineering and Industrial Services 2021 – 2022

- Led design direction for industrial machine fabrication projects, coordinating 6 team members across product, part, and component development.
- Created SolidWorks designs for full-scale industrial machine prototypes, including sheet metal, machined parts, fabricated assemblies, BOMs, and vendor-ready drawings.

SELECT PROJECTS

Witmotion BLE IMU Streaming — C, Python, Zephyr, nRF52, BLE, SEGGER RTT

- Built Zephyr firmware on an nRF52832 DK as a BLE central for a Witmotion IMU, with GATT discovery, notification handling, reconnect scanning, and packet decoding.
- Decoded accel/gyro/orientation packets with CRC16 validation; logged timestamped CSV data with sequence, RSSI, Hz, and IMU fields.

ROS2 Mobile Robot Stack — ROS2, Nav2, Gazebo, Python, Linux

- Built a differential-drive ROS2 robot model in URDF/xacro and Gazebo, maintaining a stable tf2 transform tree.
- Implemented Nav2 waypoint navigation with launch-based bring-up, planner/controller tuning, rosbag2 logging, and diagnostics.

Motor Driver PCB Design — KiCad, H-Bridge Driver, INA240, C, PID Control

- Designed a 2-layer KiCad motor controller PCB with H-bridge drive, INA240 current sensing, encoder conditioning, separated power/analog zones, and per-pin decoupling.
- Built a 100 Hz C PID position/velocity controller with hardware quadrature decoding, anti-windup, and inner current/torque loop from shunt sensing; validated against bench-supply measurements.

EDUCATION

Purdue University | M.S. Electrical and Computer Engineering (Part Time, Online)

In Progress

Northeastern University | M.S. Computer Science

May 2026

The Ohio State University | B.S. Mechanical Engineering

Dec 2020